

Online Retail Sales & Customer Insights Dashboard

1. Project Overview

The Online Retail Sales & Customer Insights Dashboard project aims to analyse transactional sales data from an online retail business to uncover valuable insights into revenue performance, customer purchasing behaviour, product performance, and geographic sales distribution. The project combines Python for data preprocessing, SQL for analytical querying, and Tableau for interactive dashboard development.

2. Problem Statement

Businesses generate large volumes of transactional data daily; however, extracting actionable insights from this data can be challenging. The objective of this project is to transform raw sales data into meaningful business intelligence that supports data-driven decision-making regarding sales growth, customer retention, and product strategy.

3. Dataset Summary

The dataset contains online retail transactions recorded between 2010 and 2011.

Dataset Features

Column Name	Description
Invoice	Unique invoice number for each transaction
StockCode	Product identification code
Description	Product description
Quantity	Number of units purchased
InvoiceDate	Date and time of transaction
Price	Unit price of the product
Customer_ID	Unique customer identifier
Country	Country where the customer is located

Feature Engineering

Additional columns were created during preprocessing:

New Column	Description
Revenue	Quantity × Price
OrderType	Categorizes orders as Retail or Wholesale
IsCancellation	Identifies cancelled/returned transactions

4. Data Cleaning and Preprocessing

The following preprocessing steps were performed using Python:

- Removed records with missing Customer IDs.
 - Converted InvoiceDate to DateTime format.
 - Created Revenue column using Quantity × Price.
 - Created IsCancellation flag based on negative quantities.
 - Converted Customer_ID from decimal format to string format.
 - Created OrderType classification.
 - Removed data quality issues and prepared the dataset for analysis.
-

5. Project Objectives

The project focuses on answering the following business questions:

1. What is the overall revenue trend month-over-month?

	Month	TotalRevenue
1	2009-12	686654.16
2	2010-01	557319.06
3	2010-02	506371.06
4	2010-03	699608.99
5	2010-04	594609.19
6	2010-05	599985.79
7	2010-06	639066.58
8	2010-07	591636.74
9	2010-08	604242.65
10	2010-09	831615.00
11	2010-10	1036680.00
12	2010-11	1172336.04
13	2010-12	884591.89
14	2011-01	569445.04
15	2011-02	447137.35
16	2011-03	595500.76

17	2011-04	469200.36
18	2011-05	678594.56
19	2011-06	661213.69
20	2011-07	600091.01
21	2011-08	645343.90
22	2011-09	952838.38
23	2011-10	1039318.79
24	2011-11	1161817.38
25	2011-12	518210.79

2. Which products generate the highest revenue and quantity sold?

	Description	TotalRevenue
1	REGENCY CAKESTAND 3 TIER	286486.30
2	WHITE HANGING HEART T-LIGHT HOLDER	252072.46
3	PAPER CRAFT , LITTLE BIRDIE	168469.60
4	Manual	152340.57
5	JUMBO BAG RED RETROSPOT	136980.08
6	ASSORTED COLOUR BIRD ORNAMENT	127074.17
7	POSTAGE	126563.04
8	PARTY BUNTING	103880.23

	Description	TotalQuantitySold
1	WORLD WAR 2 GLIDERS ASSTD DESIGNS	109169
2	WHITE HANGING HEART T-LIGHT HOLDER	93640
3	PAPER CRAFT , LITTLE BIRDIE	80995
4	ASSORTED COLOUR BIRD ORNAMENT	79913
5	MEDIUM CERAMIC TOP STORAGE JAR	77916
6	JUMBO BAG RED RETROSPOT	75759
7	BROCADE RING PURSE	71129

3. Which countries contribute the most revenue?

	Country	TotalRevenue
1	United Kingdom	14723147.50
2	EIRE	621631.11
3	Netherlands	554232.34
4	Germany	431262.46
5	France	355257.47
6	Australia	169968.11
7	Spain	109178.53
8	Switzerland	100365.34
9	Sweden	91549.72
10	Denmark	69862.19
11	Belgium	65753.42
12	Norway	60109.62

	Market	Revenue
1	Intemational	3020281.66
2	UK	14723147.50

4. What is the customer repeat-purchase rate?

	RepeatPurchaseRate
1	72.39

5. What is the Average Order Value (AOV) trend?

	Month	AverageOrderValue		Month	AverageOrderValue
1	2009-12	454.140000	13	2010-12	631.850000
2	2010-01	551.260000	14	2011-01	576.950000
3	2010-02	458.670000	15	2011-02	448.480000
4	2010-03	459.060000	16	2011-03	450.800000
5	2010-04	447.410000	17	2011-04	408.360000
6	2010-05	435.720000	18	2011-05	436.400000
7	2010-06	426.900000	19	2011-06	474.670000
8	2010-07	428.410000	20	2011-07	450.860000
9	2010-08	467.320000	21	2011-08	504.170000
10	2010-09	492.370000	22	2011-09	542.930000
11	2010-10	486.020000	23	2011-10	538.790000
12	2010-11	453.160000	24	2011-11	437.270000
			25	2011-12	666.080000

6. What percentage of revenue comes from the top 20% of customers?

	RevenueContributionPercent
1	77.260000

7. What actionable business recommendations can be derived from the analysis?

	Country	Revenue
1	United Kingdom	14723147.50
2	EIRE	621631.11
3	Netherlands	554232.34
4	Germany	431262.46
5	France	355257.47

	Customer_ID	Revenue
1	18102.0	608821.65
2	14646.0	528602.52
3	14156.0	313946.37
4	14911.0	295972.63
5	17450.0	246973.09
6	13694.0	196482.81
7	17511.0	175603.55
8	16446.0	168472.50
9	16684.0	147142.77
10	12415.0	144458.37

	Description	Revenue
1	REGENCY CAKESTAND 3 TIER	286486.30
2	WHITE HANGING HEART T-LIGHT HOLDER	252072.46
3	PAPER CRAFT , LITTLE BIRDIE	168469.60
4	Manual	152340.57
5	JUMBO BAG RED RETROSPOT	136980.08
6	ASSORTED COLOUR BIRD ORNAMENT	127074.17
7	POSTAGE	126563.04
8	PARTY BUNTING	103880.23
9	MEDIUM CERAMIC TOP STORAGE JAR	81416.73
10	PAPER CHAIN KIT 50'S CHRISTMAS	79594.33

6. Tools and Technologies Used

- Python (Pandas, NumPy)
- SQL Server
- Tableau
- Microsoft Excel
-

7. Key Performance Indicators (KPIs)

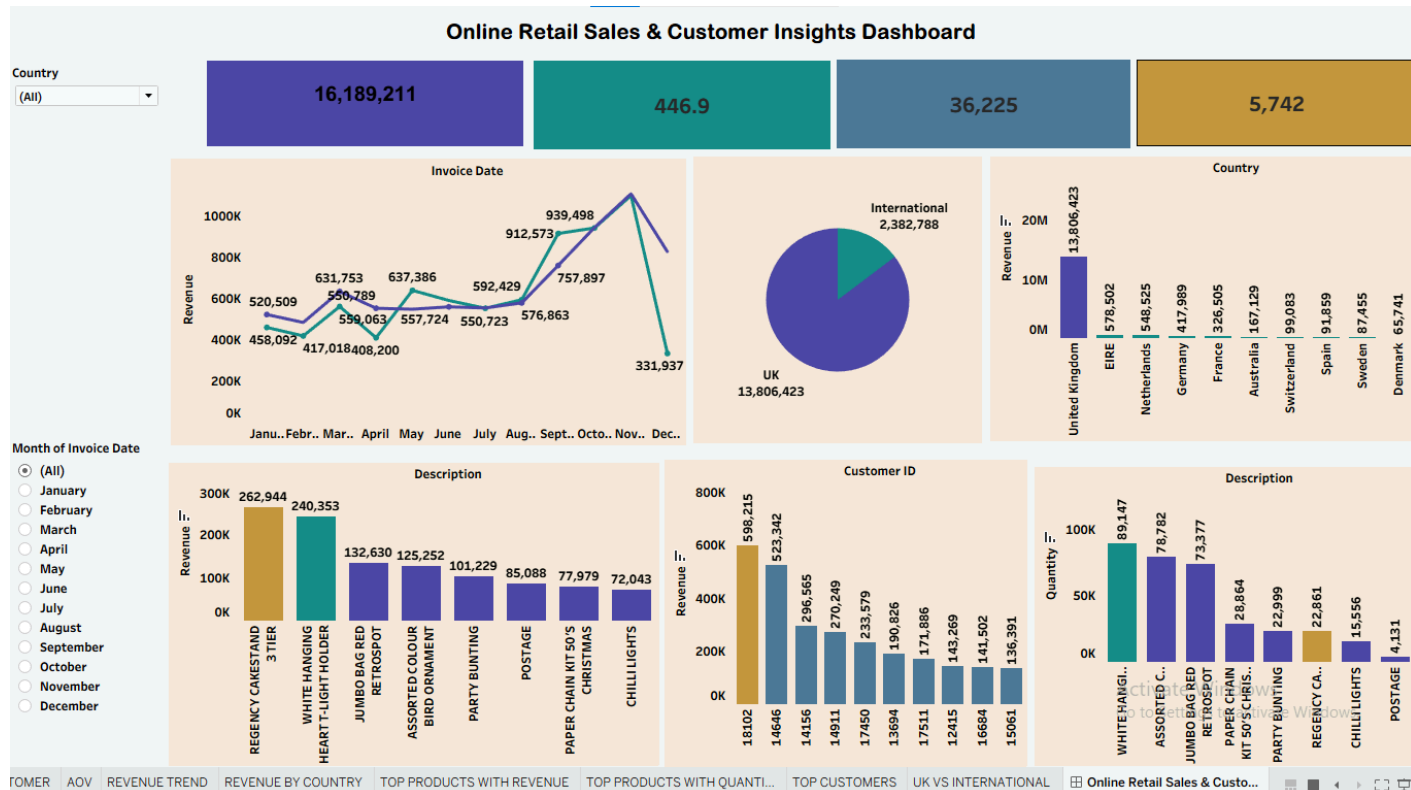
The dashboard tracks the following KPIs:

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value (AOV)

8. Dashboard Visualisations

The Tableau dashboard includes:

- Revenue Trend Analysis
- Revenue by Country
- Top 10 Products by Revenue
- Top 10 Products by Quantity Sold
- Top 10 Customers by Revenue
- UK vs International Revenue Comparison
- KPI Summary Cards



9. Key Findings

- Revenue exhibited noticeable fluctuations across different months, indicating seasonal purchasing patterns.
- A limited number of products generated a significant proportion of total revenue.
- The United Kingdom contributed the largest share of revenue, demonstrating strong market dependence.
- Repeat customers played a crucial role in overall sales performance.
- Revenue concentration analysis revealed that a relatively small percentage of customers generated a large portion of total revenue.
- Product performance varied significantly, highlighting opportunities for inventory optimisation and targeted marketing.

10. Business Recommendations

- Strengthen customer retention initiatives to increase repeat purchases.
- Promote high-performing products through targeted campaigns and bundle offers.
- Explore opportunities to expand in international markets.
- Increase Average Order Value through cross-selling and upselling strategies.
- Develop loyalty programmes for high-value customers.

- Monitor product performance regularly to optimize inventory management.
-

11. Conclusion

This project successfully transformed raw retail transaction data into actionable business insights through data cleaning, SQL analysis, and Tableau visualization. The dashboard provides stakeholders with a comprehensive view of sales performance, customer behavior, and product trends, enabling informed decision-making and strategic business planning.